Irrationality criteria for Lambert subseries

The series $\sum_{a\in A}(b^a-1)^{-1}$ and Erdős Problem #257

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Abstract

We give two hereditary irrationality criteria for Lambert subseries. If E satisfies a finite-prime weighted summability condition at base two and V admits a summable positive divisor cover, then $\sum_{a\in A}(b^a-1)^{-1}$ is irrational for every infinite $A \subseteq E \cup V$ and every integer $b \geq 2$. The two support classes are incomparable, both reach beyond reciprocal summability, and their unions can lie in neither class. The proof averages first over multiples of a modulus and then over dyadic lengths. The second average restores a reciprocal factor lost at incomplete residue periods; using the same average for both criteria gives one index at which both errors are small. Arbitrary infinite support at base two remains open.

1 Introduction

For a set A of positive integers and a real number $b > 1$, write

$$X_A(b) = \sum_{a\in A} \frac{1}{b^a-1}.$$

This series converges by comparison with a geometric series, since $(b^a-1)^{-1} \leq b^{1-a}/(b-1)$. Erdős proved irrationality for full support at integer bases [5], and later proved the same conclusion for pairwise coprime supports of finite reciprocal sum [7, p. 222]. On that page he stated that the coprimality assumption could be removed, without giving the details. The question for arbitrary infinite support in base two is Problem #257:

Problem 1.1 (Erdős #257)**.** Is $X_A(2)$ irrational for every infinite $A \subseteq \mathbb{N}_{>0}$?

We seek conditions inherited by every infinite subset of the support. The first replaces reciprocal summability by a divisibility-sensitive weight. For $a = 2^k m$ with m odd, it uses $1/[m(b^{2^k}-1)]$ in place of $1/(2^k m)$: a large power of 2 can compensate for many odd cofactors. More generally, fix a finite set P of primes and write $h(a) = \prod_{p\in P} p^{v_p(a)}$ for the *P-part* of a, where $v_p(a)$ is the exponent of p in a. The prime powers may grow with a, but the prime set is fixed.

Authorship and AI use. Will Cook built and directed the research infrastructure and reviewed the claims when he could. AI agents did most of the research and drafting. Cook did not independently verify every claim.

Theorem 1.2 (a weighted condition on the support)**.** *Let $b \geq 2$ be an integer, let $A \subseteq \mathbb{N}_{>0}$ be infinite, and let P be a finite nonempty set of primes. Set $h(a) = \prod_{p \in P} p^{v_p(a)}$. If*

$$W_{b,P}(A) := \sum_{a \in A} \frac{h(a)}{a(b^{h(a)} - 1)} < \infty, \tag{1}$$

then $X_A(b)$ is irrational. In particular, the base-two condition

$$\sum_{a \in A} \frac{h(a)}{a(2^{h(a)} - 1)} < \infty \tag{W}$$

implies that $X_A(b)$ is irrational for every integer $b \geq 2$. Both conclusions are hereditary under passage to infinite subsets.

Consider the support

$$A_\star = \{2^k m : k \geq 1,\ m \text{ odd},\ m \leq 2^{2^k}\}.$$

For a fixed k, the sum of $1/m$ over the odd cofactors is of order 2^k. Thus the corresponding terms contribute at least $1/4$ to $\sum_{a \in A_\star} 1/a$, but at most 2^{1-k} to (W). The calculation in Section 2 proves both bounds. Every infinite subset of $A_\star$ consequently gives an irrational value at every integer base. Since $h/(2^h - 1) \leq 1$, the same theorem also includes reciprocal-summable supports. Section A gives a direct proof of Erdős's stated extension.

The second condition uses joint divisibility: it bounds fractional powers of divisor counts by positive divisor sums (Theorem 3.1). Neither condition implies the other, but they combine: every infinite subset of $E \cup V$ has irrational sum when E satisfies the base-two weighted condition and V has a summable divisor cover (Theorem 4.1). Two irrational subseries could sum to a rational number, so we must control their errors at the same index. The separating constructions in Proposition 5.1 give such a union outside both individual classes (Corollary 5.2). These are sufficient conditions, not a characterisation of the supports in Problem 1.1.

The proofs compare a multiple of $X_A(b)$ with an integer. For a positive integer N, put

$$\Delta_{b,A}(N) = \sum_{a \in A} \frac{b^{N \bmod a} - 1}{b^a - 1} = (b^N - 1)X_A(b) - J_{b,A}(N), \qquad J_{b,A}(N) = \sum_{\substack{a \in A \\ a \leq N}} \sum_{j=1}^{\lfloor N/a \rfloor} b^{N-ja}. \tag{D}$$

Division of N by each a gives the identity. At an integer base, $J_{b,A}(N)$ is an integer; for infinite A, the displacement is positive because some $a \in A$ exceeds N. Thus, if $X_A(b) = p/q$ with $p \in \mathbb{Z}$ and $q \geq 1$ an integer, then $q\Delta_{b,A}(N)$ is a positive integer and $\Delta_{b,A}(N) \geq 1/q$. We seek displacements tending to zero; $\Delta_{b,A}(N)$ itself is not a fractional part and may exceed 1. Erdős also proposed using fractional parts of powers times the series value [7, p. 226].

Divisibility cancels finitely many exponents when N is a multiple of their least common multiple. The remaining terms have small complete-period means when their gcd

with the modulus is large. Incomplete-period bounds lack a factor $1/a$. Averaging over dyadic lengths restores it: an exponent enters the truncated error only when the sampled range reaches it, and the reciprocals of subsequent lengths form a geometric tail bounded by a multiple of $1/a$. Exponents beyond the sampled range are estimated separately. Explicit modulus dependence lets both criteria share this average.

Sections 2–4 prove the three criteria; Sections 5–6 separate their support classes and explain the method's limits. The appendices give the direct reciprocal-summable proof, finite-denominator results and rational-membership tests. The last of these decide neither $1/2$ nor $1/21$.

2 Finite averages and the weighted criterion

Fix the integer base b and the finite prime set P. We first estimate averages over multiples of a fixed modulus Q, then choose Q to exploit the weighted hypothesis. The role of the gcd is already visible at $b = 2$, $Q = 4$ and $a = 6$: the residues are $4, 2, 0$, and

$$\frac{1}{3}\left(\frac{2^4-1}{63}+\frac{2^2-1}{63}+0\right)=\frac{2}{21}\le\frac{2}{6(2^2-1)}=\frac{1}{9}.$$

The final expression is the mean of $2^{4m \bmod 6}/63$ over the same period. If $6 \mid Q$, the term with exponent 6 vanishes at every multiple of Q. In the proof, divisibility removes finitely many exponents, and the gcd estimate bounds the remaining terms.

Proof of Theorem 1.2. The contribution of one exponent. Put $d_a(m) = (b^{m \bmod a}-1)/(b^a-1)$ and $g = (Q, a)$. For positive integers Q, a, T, the orbit has length a/g. Splitting $1, \dots, T$ into full orbits and a remainder gives

$$\frac{1}{T}\sum_{t=1}^{T} d_a(tQ) \le \frac{g}{a(b^g-1)}+\frac{1}{T(b^g-1)}. \tag{2}$$

Dropping the -1 in the numerator gives the orbit sum $\sum_{t=1}^{a/g} b^{tQ \bmod a}/(b^a-1) = 1/(b^g-1)$. Complete orbits give the first term in (2); nonnegativity bounds the remaining indices by one more orbit. The theorem's weighted summand bounds the complete-period term whenever $g \ge h(a)$. The modulus below ensures this for $h(a) \le H$; larger prime parts will instead give a uniform lower bound on g. Also, for $Y = QT$,

$$\frac{1}{T}\sum_{t=1}^{T}\sum_{\substack{a\in A\\ a>Y}} d_a(tQ) \le \frac{4}{T}, \tag{3}$$

because $d_a(tQ) \le 2\,2^{tQ-a}$ when $a > QT$ and $\sum_{t=1}^{T} 2^{tQ-QT} \le 2$.

Recovering the reciprocal factor. The error in (2) tends to zero for each fixed a, but without a factor $1/a$ this does not control its sum over the growing range $a \le QT$. Take $T = 2^j$ and sum first over j. The truncated error includes a fixed a only when $2^j \ge a/Q$; the reciprocals of those lengths form a geometric tail of sum at most $2Q/a$. Thus, for

integers $Q, M \geq 1$ and $\alpha_a \geq 0$ with $\sum_a \alpha_a/a < \infty$,

$$\sum_{j=M}^{2M-1} \frac{1}{2^j} \sum_{a \leq Q2^j} \alpha_a \leq 2Q \sum_{a \geq 1} \frac{\alpha_a}{a}; \tag{4}$$

the estimate is unchanged if the available dyadic lengths start later. Dividing by M will turn the cost $2Q$ into $2Q/M$.

Choosing the modulus. Fix $\varepsilon > 0$. Choose finite nonempty $F \subseteq A$ so that the sum in (1) over $A \setminus F$ is below ε, and let L be any fixed positive common multiple of F. Besides cancelling F, the modulus should contain every P-prime power up to a growing threshold H. Truncating each prime power separately keeps its size polynomial in H. With $p_* = \max P$ and $H \geq 2p_*$, set

$$Q = L \prod_{p \in P} p^{\lfloor \log_p H \rfloor}, \qquad G = \left\lfloor \frac{H}{p_*} \right\rfloor.$$

For $a \in F$ the displacement term vanishes, since $a \mid Q$. For $a \notin F$ with $h(a) \leq H$, we have $h(a) \mid Q$ and hence $(Q, a) \geq h(a)$. The complete-period term in (2) is therefore bounded by $h(a)/[a(b^{h(a)} - 1)]$, using the monotonicity of $n/(b^n - 1)$. The complete-period contributions therefore sum to less than ε. The dyadic estimate will control their incomplete periods.

For $h(a) > H$, we instead have $(Q, a) \geq G$. Indeed, if every P-prime-power component is at most H, then $h(a) \mid Q$ and $(Q, a) \geq h(a) > H$; otherwise some $p^{v_p(a)} > H$ contributes $p^{\lfloor \log_p H \rfloor} > H/p \geq H/p_* \geq G$ to the gcd. Sum the complete-period bounds over $a \leq QT$, using $\sum_{a \leq QT} 1/a \leq 1 + \log(QT)$. There are at most QT incomplete-period terms, each at most $1/[T(b^G - 1)]$, so the total contribution from these exponents is at most

$$\frac{G(1 + \log(QT)) + Q}{b^G - 1}.$$

Balancing the two errors. Increasing M trades a smaller incomplete-period cost Q/M for a larger large-gcd cost GM/b^G, since $T = 2^j$ with $j < 2M$. With F and L fixed, $Q \leq LH^{|P|}$ is polynomial in H, whereas $G = H/p_* + O(1)$ is linear. Thus $M = \lfloor b^{G/2} \rfloor$ makes both costs tend to zero. Here M counts the scales $M \leq j < 2M$; their observation lengths are 2^j. Apply (4) with $\alpha_a = \mathbf{1}_A(a)/(b^{h(a)} - 1)$; since $h(a) \geq 1$, $\sum_a \alpha_a/a \leq W_{b,P}(A)$. Together with (3), the preceding estimates give

$$\frac{1}{M} \sum_{j=M}^{2M-1} \frac{1}{2^j} \sum_{t=1}^{2^j} \Delta_{b,A}(tQ) \leq \varepsilon + \frac{2QW_{b,P}(A)}{M} + \frac{G(1 + \log Q + 2M \log 2) + Q}{b^G - 1} + 4\,2^{-M}. \tag{5}$$

The first two terms control complete and incomplete periods with $h(a) \leq H$; the last two control $h(a) > H$ within $a \leq QT$ and the tail $a > QT$. With F and L fixed, every term after ε tends to zero as $H \to \infty$. The left-hand side is a probability-weighted average: choose one of the M scales uniformly, then one of its 2^j indices uniformly. For large H, at least one sampled $N = Qt$ therefore satisfies $\Delta_{b,A}(N) < 2\varepsilon$. Since ε is arbitrary, this contradicts the rational lower bound $1/q$. The argument gives no rate of decay in N. Finally $W_{b,P}(A) \leq W_{2,P}(A)$ for $b \geq 2$, and the weighted sum decreases on taking subsets. □

Duverney–Tachiya [1, Section 2, (2.3)–(2.9)] also bound an average on an arithmetic progression and select an index controlling a combined quantity. Their progression is built by the Chinese remainder theorem. Here the additional dyadic average controls the incomplete-period error as the modulus grows.

2.1 An example beyond reciprocal summability

For the support in the introduction, the two sums can be compared layer by layer. Recall that

$$A_\star = \{2^k m : k \geq 1,\ m \text{ odd},\ m \leq 2^{2^k}\}.$$

The layers are disjoint, since their elements have different 2-adic valuations. For $r \geq 2$, put

$$S_r = \sum_{\substack{m \leq 2^r \\ m \text{ odd}}} \frac{1}{m}.$$

Each block $2^j \leq m < 2^{j+1}$, $1 \leq j < r$, contains 2^{j-1} odd integers, with reciprocal sum between $1/4$ and $1/2$. Including $m = 1$ gives $r/4 \leq S_r \leq r$. The sum of reciprocals in the kth layer is $2^{-k} S_{2^k} \geq 1/4$, so $\sum_{a \in A_\star} 1/a$ diverges. For $P = \{2\}$, its contribution to the weighted sum satisfies

$$\frac{S_{2^k}}{2^{2^k} - 1} \leq \frac{2^k}{2^{2^k} - 1} \leq 2^{1-k}.$$

The last bound uses $2^{2^k} - 1 \geq 2^{2^k - 1}$ and $2^k \geq 2k$. Summation proves (W), despite the divergent reciprocal sum.

The sets of primes for which the weighted sum converges can also be prescribed. Let E be a finite set of primes and let $\mathcal{U}$ be an upward-closed family of subsets of E containing E but not $\emptyset$. A finite union of the constructions just described has divergent reciprocal sum, and, for every $b \geq 2$ and finite prime set P, its weighted sum $W_{b,P}$ is finite exactly when $P \cap E \in \mathcal{U}$. Taking $P = E$ shows that every infinite subset has an irrational sum at every integer base. The construction is given in Section 1.3 of the companion paper.

3 Positive divisor majorants

The weighted criterion treats exponents separately. A divisor cover instead uses their joint divisibility. To describe it, write the series in terms of its divisor counts:

$$c_A(n) = \#\{a \in A : a \mid n\}, \qquad X_A(b) = \sum_{n \geq 1} c_A(n) b^{-n}. \tag{6}$$

Expand $(b^a - 1)^{-1} = \sum_{j \geq 1} b^{-aj}$ and interchange the nonnegative sums. The selector $\mathbf{1}_A$ is zero or one, whereas $0 \leq c_A(n) \leq \tau(n)$, with $\tau(n)$ the number of positive divisors of n. These are power-series coefficients before any carrying; they need not be base-b digits.

Kaneko–Suzuki–Tachiya [2, Theorems 1 and 3] require sparsity of the nonzero power-series coefficients. Sparsity of the selected Lambert denominators does not suffice: for any nonempty A, $c_A(n)$ is positive on every multiple of $\min A$. Its nonzero

positions therefore have positive lower density, violating those support-counting hypotheses. The criteria do not apply directly to (6). The density calculation and the distinction between their remote-tail average and our displacement appear in Section 1.2 of the companion paper.

3.1 A finite divisor set and the cover condition

For finite $F \subseteq \mathbb{N}_{>0}$, write $f_F(n) = \#\{a \in F : a \mid n\}$. Consider the four exponents $F = \{2, 6, 10, 30\}$. If $2 \nmid n$, the count is zero. Otherwise each of the primes 3 and 5 that divides n doubles the count. Thus

$$f_F(n) = \mathbf{1}_{2|n}(1 + \mathbf{1}_{3|n})(1 + \mathbf{1}_{5|n}).$$

For $0 < \alpha \leq 1$, put $z = 2^\alpha - 1$. Taking the fractional power and expanding gives a positive divisor sum:

$$f_F(n)^\alpha = \mathbf{1}_{2|n} + z\mathbf{1}_{6|n} + z\mathbf{1}_{10|n} + z^2\mathbf{1}_{30|n}, \qquad C = \frac{1}{2}\left(1 + \frac{z}{3}\right)\left(1 + \frac{z}{5}\right).$$

Since $\mathbf{1}_{d|n}$ has mean $1/d$, C is the mean of this expansion. The same factorisation applies to $\{qd : d \mid \prod_{p\in P} p\}$ when no prime in P divides q. Small α reduces the coefficients, but the tail estimate costs $(2^\alpha - 1)^{-1} = 1/z$. The condition below balances these effects over a sequence of finite sets.

Theorem 3.1 (a summable divisor-cover criterion)**.** *For each $j \geq 1$, let $F_j \subseteq \mathbb{N}_{>0}$ be finite, let $0 < \alpha_j \leq 1$, and let $c_{j,d} \geq 0$ satisfy*

$$f_{F_j}(n)^{\alpha_j} \leq \sum_{d|n} c_{j,d} \quad (n \geq 1).$$

Set $C_j = \sum_{d\geq 1} c_{j,d}/d$. If

$$\sum_{j\geq 1} \frac{C_j 2^{j\alpha_j}}{2^{\alpha_j} - 1} < \infty, \tag{V}$$

then $X_A(b)$ is irrational for every infinite $A \subseteq \bigcup_j F_j$ and every integer $b \geq 2$.

For example, take $F_j = \{4^j\}$, $\alpha_j = 1$ and $c_{j,4^j} = 1$, with all other coefficients zero. Then $C_j = 4^{-j}$, and the cost in (V) is $\sum_j 2^{-j}$. Every finite set has the majorant $\alpha_j = 1$, $c_{j,d} = \mathbf{1}_{F_j}(d)$; the restriction is summability of the full cost. Covering sets may overlap, and their majorants may use divisors outside them. Section 5 gives a necessary condition independent of these choices.

3.2 A finite mean for shifted divisor tails

A divisor majorant becomes a sum of geometric tails, one for each divisor. Indeed, the positive offsets r with $d \mid N + r$ are $d - (N \bmod d), 2d - (N \bmod d), \ldots$, so

$$\sum_{\substack{r\geq 1 \\ d|N+r}} B^{-r} = \frac{B^{N \bmod d}}{B^d - 1}.$$

Fractional powers introduce $B = 2^\alpha$, possibly close to 1. The estimate must remain uniform after multiplication by $B-1$ and allow the modulus to grow: the mixed proof fixes the first covering sets before enlarging that modulus.

For $1 < B \le 2$, positive integers L, d, M, and an integer $R \ge 0$, put

$$w_{B,d}(n) = \frac{B^{n \bmod d}}{B^d - 1}, \qquad \mathcal{D}_{L;R,M}F = \frac{1}{M}\sum_{k=R}^{R+M-1}\frac{1}{2^k}\sum_{m=1}^{2^k} F(Lm).$$

Choose k uniformly from $R, \dots, R+M-1$, then m uniformly from $1, \dots, 2^k$ and observe $N = Lm$. Scales, not individual observations, have equal weight; repeated observations of N add their weights.

The finite estimate

$$\mathcal{D}_{L;R,M}w_{B,d} \le \frac{1 + 4L/M}{d(B-1)} \tag{S}$$

is uniform as $B \downarrow 1$ after multiplication by $B-1$: the normalised bound is $(1+4L/M)/d$. The unnormalised factor $(B-1)^{-1}$ becomes the cost $(2^\alpha - 1)^{-1}$ in (V). The bound is independent of the starting scale R and tracks the ratio L/M, which tends to zero on the weighted proof's scales.

Fix $T = 2^k$ and compare d with the largest sampled multiple LT. When $d \le LT$, the complete orbit has length d/g, with $g = (L, d)$, and total weight $1/(B^g - 1)$. Split the sample into complete cycles (possibly none) and one remaining piece to obtain

$$\frac{1}{T}\sum_{m=1}^{T} w_{B,d}(Lm) \le \frac{g}{d(B^g-1)} + \frac{1}{T(B^g-1)} \le \frac{1}{d(B-1)} + \frac{1}{T(B-1)}.$$

Across dyadic lengths satisfying $d \le L2^k$, the reciprocal-length errors sum to at most $2L/[d(B-1)]$. When $d > 2LT$, we have $Lm \bmod d = Lm$ and $2Lm \le d-1$; hence $\sum_{i=0}^{d-1} B^i \ge dB^{(d-1)/2} \ge dB^{Lm}$. Each term $w_{B,d}(Lm)$ is then at most $1/[d(B-1)]$. For fixed d, the remaining transition range $LT < d \le 2LT$ occurs at most once as T doubles. At that scale, geometric summation gives

$$\frac{1}{T}\sum_{m=1}^{T} w_{B,d}(Lm) = \frac{B^L}{T(B^L-1)}\,\frac{B^{LT}-1}{B^d-1} \le \frac{LB^L}{d(B^L-1)} \le \frac{2L}{d(B-1)}.$$

The first inequality uses the convexity of $x \mapsto B^x - 1$ and its value 0 at the origin to bound the ratio by LT/d. For the last inequality we used $(B^L-1)/(B-1) = \sum_{i=0}^{L-1} B^i \ge B^{L-1}$ and $B \le 2$. The incomplete periods and the possible transition scale together cost at most $4L/[d(B-1)]$ before division by M. Adding the mean term $1/[d(B-1)]$ proves (S). The corresponding source estimate uses the same finite average. Nonnegative interchange permits summation against any coefficients c_d with $\sum_d c_d/d < \infty$.

3.3 Proof of the cover criterion

Proof of Theorem 3.1. Write $B_j = 2^{\alpha_j}$ and

$$U_j(N) = \sum_{r\ge1} 2^{-r} f_{F_j}(N+r), \qquad V_j(N) = \sum_{d\ge1} c_{j,d} w_{B_j,d}(N).$$

The U_j are the tails to be made small; the V_j have means controlled by (S). Subadditivity of the power α_j and the divisor majorant connect them:

$$U_j(N)^{\alpha_j} \le \sum_{r\ge1} B_j^{-r} f_{F_j}(N+r)^{\alpha_j} \le \sum_{d\ge1} c_{j,d} \sum_{\substack{r\ge1\\ d\mid N+r}} B_j^{-r} = V_j(N).$$

For the displacement itself, the geometric-series identity gives

$$\Delta_{2,F_j}(N) = U_j(N) - X_{F_j}(2) \le U_j(N).$$

For $j \le J$, divisibility will cancel the displacement, so $U_j(N) = X_{F_j}(2)$. Only the tails with $j > J$ must be made small.

Fix $\varepsilon > 0$ and allot $t_j = \varepsilon 2^{-j}$ to the jth tail. These allowances sum to ε. The sufficient condition $t_j^{-\alpha_j} V_j < 1$ for $U_j < t_j$ explains the factor $2^{j\alpha_j}$ in the cover cost. Choose J with

$$K_J := \sum_{j>J} \frac{C_j t_j^{-\alpha_j}}{B_j - 1} < \frac{1}{4}.$$

This is possible since $\varepsilon^{-\alpha_j} \le \max(1, \varepsilon^{-1})$. Choose L divisible by every member of the first J finite sets. For every $R \ge 0$ and $M \ge 4L$, (S) bounds the $\mathcal{D}_{L;R,M}$-mean of $S_J(N) := \sum_{j>J} t_j^{-\alpha_j} V_j(N)$ by $(1 + 4L/M)K_J < 1/2$. Choose a sample with $S_J(N) < 1$. This controls all countably many tails at once: each nonnegative summand is below 1, hence $U_j(N) < t_j$ for every $j > J$. The first J sets contribute no displacement; later sets cover the rest of A, with overlaps only enlarging the upper bound. Thus,

$$0 < \Delta_{2,A}(N) \le \sum_{j>J} U_j(N) \le \varepsilon.$$

Equation (D) now excludes rationality at base two. For $0 \le r < d$, the function $(b^r - 1)/(b^d - 1)$ is nonincreasing on $b > 1$. For $r > 0$, cancel $b-1$ and write it as $A(b)/(A(b) + C(b))$, where $A(b) = \sum_{i<r} b^i$ and $C(b) = \sum_{r\le j<d} b^j$. Its derivative has the sign of $A'C - AC' = \sum_{i<r\le j<d} (i-j) b^{i+j-1} < 0$. Thus $\Delta_{b,A}(N) \le \Delta_{2,A}(N)$ for every real $b \ge 2$; when b is an integer, this contradicts the lower bound obtained from (D) under rationality. Any prescribed positive integer can be included as a divisor of L, so the chosen indices can also be required to be arbitrarily large. □

The same proof permits any positive weights η_j with $\sum_j \eta_j = 1$: replace $2^{j\alpha_j}$ in (V) by $\eta_j^{-\alpha_j}$ and take $t_j = \varepsilon\eta_j$. These weights allocate error allowances; they do not reweight the Lambert subseries. The stronger condition $\sum_j C_j 2^{j\alpha_j} 2^{\alpha_j}/(2^{\alpha_j}-1)^2 < \infty$ implies (V), since each of its summands is the corresponding summand of (V) multiplied by $2^{\alpha_j}/(2^{\alpha_j}-1) > 1$. This proves inclusion of the hypotheses for a given cover; after optimisation over all covers, the inclusion of support classes is strict by a corresponding source theorem.

4 A common index for the two criteria

Small displacements at unrelated indices do not control the union. Put both finite cancellation requirements in one modulus, then bound both remaining errors on the same finite distribution. The weighted proof enlarges the modulus; (S) keeps the cover estimate valid. We work at base two and compare displacements for larger integer bases.

Theorem 4.1 (mixed weighted and cover supports)**.** *Let $E, V \subseteq \mathbb{N}_{>0}$. Suppose E satisfies* (W) *for a finite nonempty prime set P, and $V \subseteq \bigcup_j F_j$ for finite sets and nonnegative majorants satisfying the hypotheses of Theorem 3.1, with either* (V) *or its positive-weight variant. Then $X_A(b)$ is irrational for every infinite $A \subseteq E \cup V$ and every integer $b \geq 2$.*

Proof. Fix the finite cancellation requirements. Fix $\varepsilon > 0$ and an integer $N_0 \geq 1$, and put $\rho = \varepsilon/3$. Use the cover notation B_j, U_j, V_j above, with $\eta_j = 2^{-j}$ in the case (V). Set $t_j = \rho\eta_j$ and choose J so that

$$K_J = \sum_{j>J} \frac{C_j t_j^{-\alpha_j}}{B_j - 1} < \frac{1}{16}.$$

Choose a finite $F \subseteq E$ so that the weighted sum over $E \setminus F$ is $\kappa < \rho/16$. Let L be a positive common multiple of N_0, all members of F, and all members of the first J finite sets. Fix J, F and L before choosing scales; later moduli must be multiples of L.

Estimate both errors on one distribution. With this finite data fixed, use the base-two choices

$$Q = L \prod_{p \in P} p^{\lfloor \log_p H \rfloor}, \qquad G = \lfloor H / \max P \rfloor, \qquad M = \lfloor 2^{G/2} \rfloor.$$

Both estimates use $\mathcal{D}_{Q;M,M}$: the scale index k runs from M to $2M - 1$, and each observation is $N = Qm$. The estimate (5) allows any common multiple of F as L, so it accommodates the cover's extra divisors. It also holds for finite or empty E: infinitude was needed only for the final positive displacement. Consequently

$$\mathcal{D}_{Q;M,M}(\Delta_{2,E}/\rho) < \frac{1}{8}$$

for sufficiently large H. For the same finite distribution, (S) and nonnegative interchange give

$$\mathcal{D}_{Q;M,M} S_J \leq (1 + 4Q/M) K_J < \frac{1}{8}, \qquad S_J = \sum_{j>J} t_j^{-\alpha_j} V_j,$$

because $Q/M \to 0$. *Choose the index only after adding the errors.* On this distribution,

$$\mathcal{D}_{Q;M,M}(\Delta_{2,E}/\rho + S_J) < \frac{1}{4}.$$

Linearity of the mean suffices; no independence assumption is needed. Choose a sampled $N = Qm$ where the nonnegative sum is less than 1. Then $\Delta_{2,E}(N) < \rho$, and every summand of $S_J(N)$ is less than 1. Since $U_j^{\alpha_j} \leq V_j$, the same choice gives $U_j(N) < t_j$ for every $j > J$. The displacement terms from the first J finite sets vanish because their exponents divide L, and hence divide N. Therefore

$$\Delta_{2,A}(N) \leq \Delta_{2,E}(N) + \Delta_{2,V}(N) < 2\rho < \varepsilon, \qquad N \geq Q \geq L \geq N_0.$$

This inequality remains valid when E and V overlap. Since A is infinite, $\Delta_{2,A}(N) > 0$. The comparison $0 < \Delta_{b,A}(N) \le \Delta_{2,A}(N)$ proved in the cover argument therefore gives arbitrarily small positive values at every integer base. Equation (D) rules out rationality. □

5 Comparison of the support classes

Let $\mathcal{W}_b$ consist of supports satisfying the weighted condition at base b for some finite nonempty prime set, and let $\mathcal{C}$ consist of supports with a cover satisfying (V). Membership requires one finite prime set or one summable cover for the whole support. The constructions below exclude every such choice.

Proposition 5.1 (incomparable support criteria)**.** *There are infinite positive supports E and V such that*

$$E \in \mathcal{W}_2, \quad E \notin \mathcal{C}, \qquad V \in \mathcal{C}, \quad V \notin \mathcal{W}_b \ (b \ge 2), \qquad \sum_{a \in V} \frac{1}{a} = \infty.$$

Both constructions use $\{qd : d \mid \prod_{p\in P} p\}$, with opposite divisibility patterns. For E, growing powers $q = 2^k$ make the weighted sum converge, while simultaneous divisors obstruct every cover. For V, disjoint prime sets force every finite-prime weighted sum to diverge, while fractional powers of the divisor counts make the cover cost summable.

5.1 A lower bound independent of the cover

To exclude every cover, we bound its cost below using only finite subsets of the support. For finite F, let $\mathbb{E}_F$ denote the uniform mean modulo $\operatorname{lcm}(F)$, with $\operatorname{lcm}(\emptyset) = 1$ and $f_\emptyset = 0$. Allow arbitrary weights $\eta_j > 0$ with $\sum_j \eta_j = 1$, so that the lower bound also survives reweighting the cover. Set

$$K = \sum_j \frac{C_j \eta_j^{-\alpha_j}}{2^{\alpha_j} - 1}, \qquad \Psi(0) = 0, \quad \Psi(t) = \inf_{0<\alpha\le 1} \frac{t^\alpha}{2^\alpha - 1} \quad (t \ge 1).$$

Here K depends on the cover, majorants, exponents and weights; Ψ minimises only over the scalar exponent α. Write $\log^+ t = \log \max\{1, t\}$. For every finite $F \subseteq \bigcup_j F_j$,

$$K \ge \mathbb{E}_F \Psi(f_F) \ge e\, \mathbb{E}_F \log^+ f_F. \tag{7}$$

Indeed, if $f_F(n) = t > 0$, coverage gives $\sum_j f_{F_j}(n) \ge t$, so some j satisfies $f_{F_j}(n) \ge \eta_j t$. Multiply its divisor majorant by $\eta_j^{-\alpha_j}/(2^{\alpha_j} - 1)$ to obtain at least $t^{\alpha_j}/(2^{\alpha_j} - 1) \ge \Psi(t)$. The index may depend on n, but the sum of all these weighted majorants bounds $\Psi(f_F(n))$ at every n. Average over $1 \le n \le X$. Each divisor indicator has mean $\lfloor X/d \rfloor / X \le 1/d$, so the weighted majorants have total mean at most K. As $X \to \infty$, the mean of $\Psi(f_F)$ tends to $\mathbb{E}_F \Psi(f_F)$, proving the first inequality. Only this finite-support function needs a period; the covering moduli need not divide $\operatorname{lcm}(F)$. The second is immediate for $t = 0, 1$; for $t > 1$, use $2^\alpha - 1 \le \alpha$ and $e^u/u \ge e$ with $u = \alpha \log t$. The scalar inequality $t^\alpha/(2^\alpha - 1) \ge e \log t$ also has a corresponding source theorem.

For $\{qd : d \mid \prod_{p\in P} p\}$, Appendix E obtains an asymptotically sharp cost in (13). The separation below needs only (7).

5.2 Two separating constructions

Proof of Proposition 5.1. The reciprocal sum over odd primes diverges, even after finitely many are removed. Otherwise $\prod_{p\ \mathrm{odd}}(1+1/p)$ would bound the sum of reciprocals of odd squarefree integers; the decomposition $n = ds^2$ would then make the odd harmonic series converge.

A weighted support with no divisor cover. The kth group of exponents will carry a factor 2^k. The event $v_2(n) = k$ has density $2^{-(k+1)}$, so a prime reciprocal sum of order 2^k will give a fixed contribution to the mean logarithmic count. Taking its leading constant below $\log 2$ leaves exponential decay in the weighted sum. Choose pairwise disjoint finite sets P_k of odd primes such that $2^k/4 \le S_k := \sum_{p\in P_k} 1/p < 2^k/4 + 1$. Put $M_k = \prod_{p\in P_k} p$ and $E = \bigcup_{k\ge 1}\{2^k d : d \mid M_k\}$. For $P = \{2\}$ the weighted sum is

$$\sum_{k\ge 1} \frac{\prod_{p\in P_k}(1+1/p)}{2^{2^k}-1} \le 2e \sum_{k\ge 1} \exp(-(\log 2 - 1/4)2^k) < \infty.$$

For $F_m = \bigcup_{k=1}^{m}\{2^k d : d \mid M_k\}$, average $\log^+ f_{F_m}$ over a period, using the disjoint events $v_2(n) = k < m$, of density $2^{-(k+1)}$. On the kth event use only the kth group, giving $f_{F_m}(n) \ge 2^{Z_k(n)}$, where $Z_k(n) = \sum_{p\in P_k} \mathbf{1}_{p\mid n}$. Odd-prime divisibility is independent of that event, so

$$\mathbb{E}_{F_m} \log^+ f_{F_m} \ge (\log 2) \sum_{k=1}^{m-1} 2^{-(k+1)} S_k \ge \frac{m-1}{8} \log 2.$$

A cover satisfying (V) would bound all these means by its fixed cost, contradicting their growth in m. Thus $E \notin \mathcal{C}$.

A divisor cover with no finite-prime weighted condition. Choose unused odd primes $q_j \ge 4^j$ and disjoint finite sets P_j of unused odd primes. Stop each P_j at the first prefix with $R_j := \prod_{p\in P_j}(1+1/p) \ge q_j$. Prime reciprocal divergence permits this, and minimality gives $q_j \le R_j < 4q_j/3$. Put $S_j = \sum_{p\in P_j} 1/p$, $M_j = \prod_{p\in P_j} p$, and $V = \bigcup_{j\ge 1} F_j$, where $F_j = \{q_j d : d \mid M_j\}$. No prime divides exponents in two different sets F_j, and

$$\sum_{a\in F_j} \frac{1}{a} = \frac{R_j}{q_j} \in [1, 4/3).$$

Thus $\sum_{a\in V} 1/a$ diverges. Given a finite set P of primes, all exponents in all but finitely many F_j are coprime to $\prod_{p\in P} p$. On each such F_j we have $h_P(a) = 1$, so its contribution to the base-b weighted sum is $R_j/[q_j(b-1)] \ge 1/(b-1)$. Hence $V \notin \mathcal{U}_b$ for every integer $b \ge 2$.

It remains to make the entire cover cost summable. The inequalities $\log(1+x) \ge x - x^2/2$ and $\sum_{p\in P_j} p^{-2} \le 1$ give $S_j \le \log R_j + 1/2 < \log q_j + 1$. Writing $z_j = 2^{\alpha_j} - 1$, the index weight and divisor product contribute $(1+z_j)^j \prod_{p\in P_j}(1+z_j/p) \le \exp(z_j(j+S_j))$. We choose $z_j(j+S_j) = 1$ to keep this factor bounded while paying the remaining cost $1/(q_j z_j)$. Set

$$z_j = (j+S_j)^{-1}, \qquad \alpha_j = \log_2(1+z_j), \qquad c_{j,q_j d} = z_j^{\omega(d)} \quad (d \mid M_j),$$

with other coefficients zero and $\omega(d)$ the number of prime factors of d. If $q_j \mid n$ and Z primes in P_j divide n, the divisor majorant equals $(1+z_j)^Z = f_{F_j}(n)^{\alpha_j}$; otherwise both sides vanish. The sum defining C_j is $C_j = q_j^{-1}\prod_{p\in P_j}(1+z_j/p)$, whence

$$\frac{C_j 2^{j\alpha_j}}{2^{\alpha_j}-1} = \frac{(1+z_j)^j}{q_j z_j}\prod_{p\in P_j}(1+z_j/p) \le \frac{e(j+S_j)}{q_j} \le e((1+\log 4)j+1)4^{-j}.$$

The last bound uses that $(j+\log x+1)/x$ decreases for $x \ge 4^j$. The bound is summable in j and includes the factor $2^{j\alpha_j}$ required by (V); hence $V \in \mathcal{C}$. □

5.3 A strict enlargement and finite unions

The mixed theorem now applies to a support outside both individual classes.

Corollary 5.2 (a support requiring the mixed criterion)**.** *There is an infinite positive support U with $U \notin \mathcal{C}$ and $U \notin \mathcal{W}_b$ for every integer $b \ge 2$, such that $X_A(b)$ is irrational for every infinite $A \subseteq U$ and every integer $b \ge 2$.*

Proof. Let $U = E \cup V$, with E, V as in Proposition 5.1. A cover of U is also a cover of E, while convergence of the weighted sum over U would imply convergence of the sum over V for the same prime set, since all summands are nonnegative. Thus U belongs to neither individual class. Theorem 4.1 gives irrationality for every infinite subset of U at every integer base $b \ge 2$. □

The fixed-dyadic cover class in (V) is closed under finite unions. When interleaving two covers, we must control the index-dependent factor $2^{j\alpha_j}$. Place their jth finite sets at positions $2j-1, 2j$ and halve each exponent; this factor then does not increase. The old majorant coefficients still work, since $f^{\alpha/2} \le f^{\alpha}$ for every nonnegative integer f. It remains to compare the denominators in the cost. Writing $T_j = C_j 2^{j\alpha_j}/(2^{\alpha_j}-1)$ for the original jth cost, its new cost is at most

$$\frac{C_j 2^{j\alpha_j}}{2^{\alpha_j/2}-1} = (1+2^{\alpha_j/2})T_j \le (1+\sqrt{2})T_j.$$

For r covers, interleave at positions at most rj and divide each exponent by r; the cost ratio is $(2^{\alpha}-1)/(2^{\alpha/r}-1) < 2r$. Weighted supports are also closed under finite unions: take the union of their prime sets: $h_P(a)$ can only increase, and $h/(b^h-1)$ decreases with h. Both criteria are hereditary and admit finite supports. Their mixed class is therefore closed under finite unions and finite changes, already with the fixed dyadic weights in (V). Countable unions need not remain in the mixed class: every prime singleton is admitted, but full prime support fails the small-displacement condition, as shown in Section 6. The reciprocal-summable class is contained in the weighted class, since $h/(2^h-1) \le 1$. A direct proof appears in Appendix A.

6 Limits of the small-displacement criterion

Small displacement is a sufficient condition for irrationality, not a necessary one. Full support already shows the distinction:

$$\Delta_{2,\mathbb{N}_{>0}}(N) > (2^N - 1)\sum_{a>N} 2^{-a} = 1 - 2^{-N} \geq \frac{1}{2} \qquad (N \geq 1).$$

The full-support value is nevertheless irrational [5]. Thus a proof for arbitrary support cannot rely on making these displacements tend to zero in every case. Other uses of the displacement, integer linear forms and averaging arguments are not excluded.

Full prime support gives a second obstruction. Although every prime singleton is admitted by both support criteria, for the set of all primes $\mathbb{P}$ one has $\Delta_{2,\mathbb{P}}(N) > 1/3$ for every $N \geq 1$: indeed $\sum_{r\geq 1} 2^{-r}\omega(N+r) \geq 1$, whereas $X_{\mathbb{P}}(2) \leq \sum_{a\geq 2}(2^a - 1)^{-1} < 2/3$. Here $\omega(n)$ is the number of distinct prime divisors of n.

Yet Tao–Teräväinen prove the full-prime case at base 2 [3, Theorem 1.3, p. 4]. The paragraph following it states extensions to prime support at every integer base and to full prime-power support at base 2, leaving the modifications to the reader. It does not treat arbitrary infinite thinnings. The proposed thinning extension is not a premise of any theorem here.

The logarithmic lower bound is not a converse: replacing a fractional majorant by a logarithmic one need not control averages along the multiples of a prescribed modulus. The counterexample and the bound that remains valid on ordinary initial intervals are proved in Section 13 of the companion paper; those proofs were first worked out in an AI-assisted note of 17 September 2026. These are statements about finite averages; they do not rule out an irrationality criterion based on other uses of logarithmic divisor counts.

For squarefree support, Duverney and Tachiya's theorem already gives irrationality. It holds at every base 2^j, $j \geq 1$, by Corollary 1.2 and Example 1.1 of Duverney and Tachiya [1, p. 4]. Their corollary covers the s-free products of any pairwise coprime sequence of polynomial growth, and they present it as support for the conjecture of Erdős and Graham stated here as Problem 1.1. For the squarefree support with 1 removed, the divisor count is $2^{\omega(n)} - 1$, which is odd for $n \geq 2$. This excludes a first-block-divisibility condition at even bases. Adjoining 1 removes that parity obstruction while changing the sum by a rational number. The argument concerns that normalisation of the certificate, and gives no obstruction to irrationality of the sum. Details and the other known support families appear in the comparison with known support theorems in the companion paper.

A growth-only criterion gives a different class: if $A = \{c_1 < c_2 < \cdots\}$ and $\limsup_n c_n/2^n = \infty$, then $X_A(b)$ is irrational for every integer $b \geq 2$ [8, Theorem 1]. The same section of the companion paper proves the translation from denominator growth and gives examples showing that this class and reciprocal summability are incomparable. The interval-filling constructions in [9] and the freely chosen lacunary denominators of [16] are not constructions of subseries of these fixed Mersenne weights.

Prime-incidence arguments require additional care as well: multiplying all prime exponents by 2 creates positive correlations between their divisibility indicators. The

identity and covariance calculation appear at the end of the companion paper's discussion of mixed supports. Size conditions alone do not supply that independence.

6.1 Algebraic bases on separated supports

The companion *Reading eight Erdős problems together*, subsection "Divisibility cuts at every algebraic base", states a transcendence extension based on the number-field Subspace Theorem. Its hypothesis is an infinite set H with indices $L_j < M_j$, where $L_j \to \infty$ and $M_j - L_j \to \infty$, such that every $n \in H$ with $n \le L_j$ divides L_j, and every $n \in H$ with $n > L_j$ is a multiple of M_j. The claimed conclusion is transcendence of $\sum_{n\in B} w_n/(t^n-1)$ for every infinite $B \subseteq H$, bounded positive integer weights w_n and real algebraic $t > 1$.

Every divisibility chain has such cuts, as does $H_* = \bigcup_j N_j\{1, \dots, 2^{N_j}\}$ with $N_0 = 1$ and $N_{j+1} = 2N_j \operatorname{lcm}(1, \dots, 2^{N_j})$. Each block has a reciprocal sum of at least $1/2$ and contains antichains of unbounded size. The weighted sum for one prime is at most $2t^2/(t-1)^3$ at every real $t > 1$. This connects the example with the weighted criterion, although the separation condition is not asserted for every support satisfying that criterion.

The companion proof is outside this paper and is not a premise of the weighted, cover or mixed theorem. It has two AI proof reviews, no Lean proof of transcendence and no independent human review. No historical priority is asserted for the extension. The case of arbitrary infinite support at base two remains unresolved.

A Reciprocal-summable supports

Erdős stated the following extension of his pairwise-coprime theorem [7, p. 222]. It follows from Theorem 1.2, since $h/(2^h-1) \le 1$, but the direct argument explains why reciprocal summability makes the averaging simpler.

Theorem A.1 (reciprocal-summable supports)**.** *Let $A \subseteq \mathbb{N}_{>0}$ be infinite. If*

$$\sum_{a\in A} \frac{1}{a} < \infty,$$

then $X_A(b)$ is irrational for every integer $b \ge 2$.

Erdős also discussed weaker conditions [7, pp. 222, 226]. We do not identify our weighted hypothesis with the conditions he suggested.

Reciprocal summability permits two successive limits: first the observation length grows with the modulus fixed, then the modulus grows until each fixed exponent divides it. The two interchanges use different summable bounds; neither requires a period for the infinite support.

Fix the integer base $b \ge 2$ and put

$$w_{b,d}(N) = \frac{b^{N \bmod d}}{b^d - 1}, \qquad T_N^{(b)} = \sum_{d\in A} w_{b,d}(N).$$

Then $T_N^{(b)} - T_0^{(b)} = \Delta_{b,A}(N)$. For a fixed positive integer Q, the orbit of Q modulo d consists of d/g residues, where $g = (Q, d)$. Summing the geometric progression gives

$$M_{Q,b}(d) := \lim_{X\to\infty} \frac{1}{X} \sum_{m=1}^{X} w_{b,d}(Qm) = \frac{g}{d(b^g - 1)} \le \frac{1}{d}. \tag{8}$$

The example $b = 2$, $Q = 4$, $d = 6$ in Section 2 has mean $1/9$ for $w_{2,6}$, but $2/21$ for its displacement. When $d \mid Q$, $w_{b,d}(Qm) = w_{b,d}(0)$ and the displacement vanishes.

To interchange the sum over exponents and this limit, we use a bound summable in d for the fixed modulus Q. Counting positive multiples of d gives

$$\begin{aligned}\frac{1}{X}\sum_{m=1}^{X} w_{b,d}(Qm) &= \sum_{r\ge 1} b^{-r} \frac{\#\{1 \le m \le X : d \mid Qm + r\}}{X} \\ &\le \sum_{r\ge 1} b^{-r} \frac{Q + r}{d} = \frac{Q/(b-1) + b/(b-1)^2}{d} \le \frac{Q+2}{d}.\end{aligned}$$

Indeed, the counted multiples are distinct and at most $QX + r$. For this fixed Q, the majorant $(Q + 2)/d$ is summable over A. Dominated convergence therefore takes the observation-length limit inside the sum over d, giving the Cesàro mean $\sum_{d\in A} M_{Q,b}(d)$. This majorant is not uniform in growing Q.

Next let $Q_t = \operatorname{lcm}(1, \dots, t)$. For each fixed d, eventually $d \mid Q_t$ and $M_{Q_t,b}(d) = (b^d - 1)^{-1}$. A second dominated-convergence argument, using $M_{Q_t,b}(d) \le 1/d$, yields

$$\sum_{d\in A} M_{Q_t,b}(d) \longrightarrow X_A(b). \tag{9}$$

The limiting displacement means therefore tend to zero. Choose t, then a sufficiently long finite average, then a term no larger than that average. The resulting arbitrarily small positive displacements contradict the rationality bound $1/q$ in (D). This proves Theorem A.1 at every integer base.

Powerful integers (all prime exponents at least two) have the form u^2v^3, so their reciprocal sum is at most $\zeta(2)\zeta(3)$. The theorem therefore applies to every infinite subset. For full perfect-power supports, compare Duverney–Tachiya [1, Corollary 1.2, p. 4] and Luca–Tachiya's earlier linear-independence results [13].

B Finite-support denominator periods

The least common multiple of the selected exponents survives cancellation as the multiplicative order of the base modulo the reduced denominator.

Theorem B.1 (the exact denominator period)**.** *Let $F \subseteq \mathbb{N}_{>0}$ be finite and nonempty, let $b \ge 2$ be an integer, and let $D_F > 0$ be the denominator of $X_F(b)$ in lowest terms. Then D_F is coprime to b, and*

$$\operatorname{ord}_{D_F}(b) = \operatorname{lcm}\{n : n \in F\}.$$

If moreover $\operatorname{lcm}(F) \ge 2$, *then* $\operatorname{lcm}(F) < D_F$. *We use* $\operatorname{ord}_1(b) = 1$, *so the statement includes* $F = \{1\}$ *at* $b = 2$.

Put $L = \operatorname{lcm}(F)$. Clearing denominators gives $D_F \mid b^L - 1$, so the upper divisibility for the order is immediate. For the reverse, fix a divisibility-maximal $n \in F$ with $n \geq 2$ and choose a prime $\ell \mid \Phi_n(b)$. With $e = v_\ell(b^n - 1)$, the cyclotomic fact in Section E gives $\operatorname{ord}_{\ell^e}(b) = n$ for the full prime power ℓ^e. Every other selected exponent m has $n \nmid m$, hence $v_\ell(b^m - 1) < e$. The nth summand has uniquely smallest ℓ-adic valuation and cannot cancel. Thus $\ell^e \mid D_F$ and $n \mid \operatorname{ord}_{D_F}(b)$. Taking all maximal selected exponents proves the order statement; the size bound follows from $\operatorname{ord}_{D_F}(b) \mid \varphi(D_F) < D_F$ when $L \geq 2$. The case $F = \{1\}$ has order one directly.

The unique-valuation argument also permits signs ± 1 on the finite summands. This signed extension is not included in the formal statement. The distinction between primes and prime powers is visible in

$$X_{\{2,3\}}(2) = \frac{10}{21}, \qquad X_{\{2,6\}}(2) = \frac{22}{63}.$$

In both examples 2 has multiplicative order six modulo the denominator. The first combines orders two and three; the second retains the order-six prime power 9, although no prime divisor of 63 has order six. The boundary $F = \{1\}$ at base two has $D_F = L = 1$. These lower denominator bounds give no upper height control for infinite partial sums and do not decide an infinite-support value. Van Assche instead constructs approximants for the full Lambert series and proves both nonvanishing and decay of the associated integer linear forms [14, Lemma 1 and (34)–(35)]. That approximation mechanism does not follow from denominator survival alone.

In particular, $1/21$ is not a finite subseries sum. If it were, the order of 2 modulo 21 would force every selected exponent to divide 6. The exponent 1 is excluded since its weight is 1. Any sum containing 2 or 3 exceeds $1/21$, whereas the only nonzero remaining sum is $1/63$, from the exponent 6. This will distinguish finite and infinite representations in the next appendix.

C Rational membership

The preceding results start from a support and prove irrationality. We now fix a target and ask whether any support represents it. The first recurrence is a necessary condition on a rational value; the greedy recurrence later gives an exact membership criterion. Neither supplies the missing arithmetic information for $1/2$ or $1/21$.

C.1 Rational values and scaled tails

Suppose $X_A(2) = p/v$, where $p \in \mathbb{Z}$ and $v \geq 1$ is an integer. Multiplying the coefficient-series identity by $v2^N$ gives

$$z_N := v \sum_{r \geq 1} c_A(N + r)2^{-r} = 2^N p - v \sum_{n=1}^{N} c_A(n)2^{N-n} \in \mathbb{Z}.$$

Separating the first term of the tail then gives $z_{N+1} = 2z_N - vc_A(N + 1)$. All the coefficients $c_A(n)$ are divisor counts of the same set A.

For nonempty A, every $a_0 = \min A$ consecutive integers contain a multiple of a_0, so zero runs of c_A have length at most $a_0 - 1$. For infinite A, choose k exponents and a

common multiple L. Then $c_A(L) \geq k$, and the scaled tail satisfies $\sum_{r\geq 1} c_A(L-1+r)2^{-r} \geq k/2$. Thus scaled tails are unbounded.

The compatibility condition on the coefficients is expressed by Dirichlet convolution: $\mu * c_A = \mathbf{1}_A$, where μ is the Möbius function. A putative recurrence with integral forcing must therefore satisfy $(\mu * c_A)(n) \in \{0, 1\}$ for all n. The telescoping proof of the integer-recurrence criterion and the Möbius inversion argument are given under Bounds for a general coefficient sequence in the companion paper. The example $A = \{2\}$ there has value $1/3$. The correspondence permits finite supports; infinitude remains a separate requirement for a counterexample to Problem 1.1.

C.2 The achievement set

Let $w_n = (2^n - 1)^{-1}$ and

$$\mathcal{A} = \left\{ \sum_{n\geq 1} \varepsilon_n w_n : \varepsilon_n \in \{0, 1\} \right\}, \qquad R_N = \sum_{n>N} w_n.$$

The estimate

$$2^{-N} < R_N \leq 2^{-N} + \frac{2}{3} 4^{-N} < w_N \qquad (N \geq 1)$$

follows by writing $w_n = 2^{-n} + 4^{-n}/(1 - 2^{-n})$ and using $n \geq N + 1 \geq 2$ for the middle bound. If two selectors first differ at n, their sums differ by at least $w_n - R_n > 0$, proving uniqueness. Since every weight exceeds its tail, Hornich's theorem [15], as proved by Nitecki [10, Theorem 4(1), p. 9], shows that the coding image is a Cantor set of measure $\lim_N 2^N R_N$; the estimate above gives $2^N R_N \to 1$, so $\lambda(\mathcal{A}) = 1$. Concretely, fixing the first N digits gives 2^N disjoint closed intervals of length R_N. These interval unions decrease to $\mathcal{A}$, and continuity of Lebesgue measure from above gives the same limit. Kovač–Tao record this strict-tail inequality and the Cantor conclusion in the fixed-base setting [4, Remark 4.1]. The geometry section of the companion paper also treats restricted supports. For any allowed set J, removing exponents can only decrease the tail of each retained weight, so the same inequality gives injectivity. If J omits exactly the finite set F, its achievement set has measure $2^{-|F|}$; if J omits infinitely many exponents, that measure is zero. The tail comparison, injectivity and measure dichotomy have separate source statements.

C.3 The greedy recurrence

For $x \geq 0$, the greedy rule starts with $r_0 = x$ and, at rank $n \geq 1$, selects n when $r_{n-1} \geq (2^n - 1)^{-1}$, subtracting that weight if selected. Let A_x be the resulting support and $c_x(n) = \#\{a \in A_x : a \mid n\}$. Set

$$P_0 = 0, \qquad P_{N+1} = 2P_N + c_x(N + 1), \qquad Q_N = \lfloor 2^N x \rfloor - P_N.$$

Here $P_N = \sum_{j=1}^{N} 2^{N-j} c_x(j)$ truncates the coefficient series (6), not the selected Lambert weights. Thus Q_N measures coefficient truncation, whereas r_N is the real remainder after N greedy decisions. The bound $P_N \leq 2^N X_{A_x}(2) \leq 2^N x$ gives $Q_N \geq 0$, and

$$Q_{N+1} = 2Q_N + \beta_N - c_x(N + 1), \qquad \beta_N = \lfloor 2^{N+1} x \rfloor - 2\lfloor 2^N x \rfloor \in \{0, 1\}. \tag{10}$$

The recurrence is integral for every real target. Write $\delta = x - X_{A_x}(2) \geq 0$. The coefficient expansion gives

$$Q_N = \left\lfloor 2^N \delta + \sum_{r \geq 1} c_x(N+r) 2^{-r} \right\rfloor.$$

For a represented target, $\delta = 0$ and $Q_N = O(\sqrt{N})$, leaving too little room for unit increases throughout a long window. For an unrepresented target, $2^N \delta$ eventually dominates the divisor count in the recurrence and forces strict increase.

Theorem C.1 (membership and nonincreasing integer remainders)**.** *For every real $x \geq 0$, the following are equivalent:*

$$x \in \mathcal{A};$$
$$\forall K \geq 0 \; \exists N \geq K : \quad Q_{N+1} \leq Q_N;$$
$$\forall K \geq 0 \; \exists N \in [K, K + 2\lfloor \sqrt{K} \rfloor + 12) : \quad Q_{N+1} \leq Q_N.$$

Proof. For a represented target, the remaining sum is at least w_n exactly when n is selected, since $R_n < w_n$. Hence the greedy rule recovers the support, and

$$0 \leq Q_N \leq \sum_{r \geq 1} c_x(N+r) 2^{-r} \leq 2\sqrt{N} + 4,$$

since $c_x(n) \leq \tau(n) \leq 2\sqrt{n}$ and $\sqrt{N+r} \leq \sqrt{N} + \sqrt{r} \leq \sqrt{N} + r$. Put $s = \lfloor \sqrt{K} \rfloor$ and $T = 2s + 12$. Strict increase at all T steps would imply $Q_{K+T} \geq T$. But $K + T < (s+4)^2$ gives $Q_{K+T} < 2s + 12 = T$. This proves the window condition and hence cofinal nonincreases.

Conversely, if $x \notin \mathcal{A}$, then $\delta > 0$ and $Q_N \geq 2^N \delta - 1$, which eventually exceeds $c_x(N+1) \leq N + 1$. Equation (10) then makes Q_N strictly increasing, contradicting cofinal nonincreases. □

The nonincreases must occur along the target's own greedy sequence. For $x = 0$, no exponent is selected and $Q_N = 0$ at every rank. A finite subseries sum is likewise represented and satisfies the criterion. The target $x = 3/4$ lies strictly between $R_1 < 2/3$ and $w_1 = 1$, so it is not represented and its integer remainders eventually increase strictly. For $1/2$ and $1/21$, the theorem does not establish which alternative occurs.

An ordinary extension gives a uniform subpower window: for each $0 < \varepsilon < 1$, its length is $\lceil C_\varepsilon (K+1)^\varepsilon \rceil$ for a suitable constant C_ε. The linked formal proof establishes the square-root window. Indeed, $\tau(n) \leq A_\varepsilon n^\varepsilon$ gives $Q_N \leq D_\varepsilon (N+1)^\varepsilon$ for represented targets, where $D_\varepsilon = A_\varepsilon \sum_{r \geq 1} r^\varepsilon 2^{-r}$. Choose $C_\varepsilon > D_\varepsilon (C_\varepsilon + 3)^\varepsilon$, which is possible because $\varepsilon < 1$. For $T = \lceil C_\varepsilon (K+1)^\varepsilon \rceil$, strict increase at all T steps would give

$$T \leq Q_{K+T} \leq D_\varepsilon (K + T + 1)^\varepsilon \leq D_\varepsilon (C_\varepsilon + 3)^\varepsilon (K+1)^\varepsilon < T,$$

a contradiction. The converse still follows from exponential growth when x is not represented. The shorter window does not decide membership of a specified target.

At $x = 1/2$, the digits β_N vanish for $N \geq 1$; at $x = 1/21$ they are six-periodic. Thus the unresolved arithmetic input is

$$\forall K \; \exists N \geq K : \qquad c_x(N+1) \geq Q_N + \beta_N, \qquad x \in \{1/2, 1/21\}. \tag{11}$$

Both Q_N and c_x must arise from the same greedy support A_x. Counterexamples to fixed-multiplier schedules, with their finite phase masks, are retained in the companion paper. They concern those schedules; they do not decide either target.

D Integer quotients, approximation and further questions

For $1/21$ we compare integer greedy remainders with the real greedy sequence; for $1/2$ we allow arbitrary finite approximating supports. Both tests require suitable data at unbounded depths.

Integer quotients for $1/21$**.** For $M, d \geq 1$, set

$$q_M(d) = \left\lfloor \frac{2^M}{2^d - 1} \right\rfloor, \qquad T_M = \left\lfloor \frac{2^M}{21} \right\rfloor.$$

For $d \geq 2$, the geometric expansion gives $q_M(d) = \sum_{j=1}^{\lfloor M/d \rfloor} 2^{M-jd}$: the omitted fraction $2^{M \bmod d}/(2^d - 1)$ is less than 1. Thus $q_M(d)$ is the coefficient prefix contributed by exponent d. Starting with remainder T_{2R}, consider $d = 2, \dots, R$ in order and subtract $q_{2R}(d)$ whenever it does not exceed the current remainder. Let D_R be the set of selected exponents and s_R the final remainder. Thus

$$s_R = T_{2R} - \sum_{d \in D_R} q_{2R}(d) \geq 0.$$

The precision is 4^R, but D_R records decisions only through rank R. The agreement condition below also uses a second run at that same precision, continued through rank $2R$. For example, $R = 6$ gives $T_{12} = 195$, $D_6 = \{5\}$ and $s_6 = 195 - \lfloor 4096/31 \rfloor = 63 \leq 64$. This is one row satisfying the bound below, not evidence that such rows occur at unbounded depths.

Write $r_n = r_n(1/21)$ for the real greedy remainder and $A_{1/21}$ for its support. The condition $\mathcal{F}_{21}$ means that there exist integers $n, R_0, K_0 \geq 0$ with all of the following properties:

1. $r_n > R_n$, the set of positive exponents outside $A_{1/21}$ is finite, and every exponent greater than n belongs to $A_{1/21}$;
2. for every $R \geq R_0$, the integer rule with target T_{2R} and weights $q_{2R}(2), \dots, q_{2R}(2R)$ selects exactly the same exponents as the real greedy rule on $\{2, \dots, 2R\}$;
3. for every $K \geq K_0$, the interval $(K, 2K]$ contains an exponent of $A_{1/21}$.

A positive limiting remainder gives a tail violation and eventual selection of every exponent, hence the first and last clauses. The middle clause additionally requires agreement with the integer rules.

Theorem D.1 (integer-quotient tests for $1/21$)**.** *The following statements hold.*

1. $1/21 \in \mathcal{A}$ *if and only if* $\mathcal{F}_{21}$ *does not hold.*
2. *If there is an unbounded sequence of ranks* R *with* $s_R \leq 2^R$*, then* $1/21 \in \mathcal{A}$*.*

3. *On $\mathcal{F}_{21}$, eventually $s_R > 2^R$, the boundary rank $R+1$ belongs to D_{R+1}, and, for all sufficiently large R,*

$$\begin{aligned} D_{R+1} &= D_R \cup \{R+1\}, \\ s_{R+1} &= 4s_R + \left\lfloor \frac{4(2^{2R} \bmod 21)}{21} \right\rfloor - 2c_{D_R}(2R+1) - c_{D_R}(2R+2) \\ &\quad - (2^{R+1}+1). \end{aligned}$$

Here $c_{D_R}(m) = \#\{d \in D_R : d \mid m\}$.

Proof. Write $x = 1/21$, $G = A_{1/21}$ and $\delta = x - X_G(2) \geq 0$. If $x \notin \mathcal{A}$, then $\delta > 0$. The real remainder tends to δ, so every sufficiently small weight is selected and eventually $r_n > R_n$. For the integer comparisons, first choose d_0 with $w_d < \delta/2$ for $d > d_0$. Comparisons through d_0 are strict, since equality would give a finite representation of x. After division by 4^R, each quotient weight differs from w_d by less than 4^{-R}, and the remainders differ by at most $d/4^R$ whenever earlier decisions agree. For large R, the first d_0 decisions therefore agree. At later ranks $d \leq 2R$, the normalised remainder exceeds $\delta/2 > w_d$, since $2R/4^R < \delta/2$; both rules select every remaining rank through $2R$. This proves $\mathcal{F}_{21}$. A represented target has $r_n \leq R_n$ at every rank, giving the converse in (1).

For every R, the rounding errors satisfy

$$\left| x - X_{D_R}(2) - \frac{s_R}{4^R} \right| \leq \frac{R+1}{4^R}. \tag{12}$$

Thus $s_R \leq 2^R$ at unbounded ranks gives finite subseries sums tending to x. Closedness of $\mathcal{A}$ proves (2), without compatibility between the sets D_R; Section B excludes a finite representation.

On $\mathcal{F}_{21}$, the eventual agreement gives $D_R = G \cap \{2, \dots, R\}$ and $D_{R+1} = D_R \cup \{R+1\}$. The rounding bound implies $s_R/4^R \to \delta > 0$. Finally, for $d \in D_R$ (so $d \geq 2$), use $q_{M+2}(d) = 4q_M(d) + 21_{d|M+1} + \mathbf{1}_{d|M+2}$, together with the analogous target identity and $q_{2R+2}(R+1) = 2^{R+1}+1$, to obtain (3). The full comparison induction and both quotient identities are given in the companion paper's integer-quotient proof for 1/21. □

In the list $q_{2R}(2), \dots, q_{2R}(R)$, each weight exceeds the sum of its successors by at least $2^R + 1$. Distinct subset sums are therefore separated by more than 2^R, and the greedy sum is the largest not exceeding the target. Thus, if $D \subseteq \{2, \dots, R\}$ and an integer s satisfy $\sum_{d \in D} q_{2R}(d) + s = T_{2R}$ with $0 \leq s \leq 2^R$, then $D = D_R$ and $s = s_R$ (uniqueness of a finite representation with the stated remainder bound). The cited finite crossing lemmas give additional consequences under their alignment hypotheses: an earlier finite prefix cannot occur, and a real greedy exponent must be skipped (the missing-prefix consequence of an aligned crossing, the real greedy skip forced by that crossing).

Approximation to $1/2$ by finite supports. For $A \subseteq \{2, \dots, M\}$ and $1 \leq m \leq M$, define the integer

$$K_A(m) = 2^{m-1} - \sum_{j=2}^{m} 2^{m-j} c_A(j).$$

It measures the error in the truncated divisor-coefficient sum, after multiplication by 2^m. In particular, $K_A(1) = 1$ and $K_A(m+1) = 2K_A(m) - c_A(m+1)$.

Theorem D.2 (finite approximations with vanishing scaled error)**.** *Suppose there are integers $M_j \geq 1$ tending to infinity and sets $A_j \subseteq \{2, \dots, M_j\}$ such that*

$$\frac{|K_{A_j}(M_j)|}{2^{M_j}} \longrightarrow 0.$$

Then $X_A(2) = 1/2$ for some infinite set $A \subseteq \mathbb{N}_{>0}$.

Only the terminal coefficient error needs control: the omitted tail has the uniform bound below. Neither agreement between the supports A_j nor bounds on earlier values of K_{A_j} are required:

$$\left| X_{A_j}(2) - \frac{1}{2} \right| \leq \frac{|K_{A_j}(M_j)| + 2\sqrt{M_j} + 4}{2^{M_j}}.$$

Indeed, $c_{A_j}(n) \leq \tau(n) \leq 2\sqrt{n}$ and $\sqrt{M+r} \leq \sqrt{M} + r$ bound the normalised tail by $\sum_{r\geq 1} 2^{-r}(2\sqrt{M}+2r) = 2\sqrt{M}+4$. Thus $X_{A_j}(2) \to 1/2$. Closedness of $\mathcal{A}$ gives a representing support; the odd reduced denominators of finite sums make it infinite. An $O(\sqrt{M_j})$ terminal error suffices, but any $o(2^{M_j})$ error is allowed. For a finite $D \subseteq \{2, \dots, M\}$, the floor-quotient identity gives

$$K_D(M) = 2^{M-1} - \sum_{d\in D} \left\lfloor \frac{2^M}{2^d - 1} \right\rfloor.$$

The exact quotient rows considered in the companion paper satisfy $\sum_{d\in D} q_M(d) = 2^{M-1} - 1$, so they have $K_D(M) = 1$. The terminal criterion allows a larger error of either sign. At a fixed depth these are different conditions, although cofinal existence of either kind of approximation characterises $1/2 \in \mathcal{A}$.

Conversely, if $X_A(2) = 1/2$, then $1 \notin A$, and the prefixes $A_M = A \cap \{2, \dots, M\}$ satisfy

$$K_{A_M}(M) = \sum_{r\geq 1} c_A(M+r)2^{-r} \leq 2\sqrt{M} + 4.$$

The right-hand side uses the full support A: truncation leaves its coefficients through M unchanged. Thus the existence hypothesis of Theorem D.2 is equivalent to $1/2 \in \mathcal{A}$. Existence at unbounded depths remains unproved; establishing it would refute Problem 1.1.

Finite families with a shared prefix. Put $B(m) = 2\lfloor \sqrt{m} \rfloor + 4$. For $0 \leq K \leq M$, consider a family $A_1, \dots, A_{B(M)} \subseteq \{2, \dots, M\}$ with

$$1 \leq K_{A_k}(m) \leq B(m) \quad (1 \leq m \leq M), \qquad K_{A_k}(M) = k.$$

Require that all the A_k agree on $\{1, \dots, K\}$ and that some integer $E \geq B(M)$ satisfies

$$\sum_{j=K+1}^{M} 1_{A_k}(j)\, 2^{M-j} + k = E \qquad (1 \leq k \leq B(M)).$$

Thus the binary suffixes have consecutive values $E - k$. The hypothesis asks for a whole family at each available depth, not merely one approximant.

Theorem D.3 (unbounded shared-prefix families represent one half)**.** *Suppose that for every N there are M, K with $\max\{N, 1\} \le M$, $0 \le K \le M$, and a family satisfying all the conditions in the preceding paragraph. Then $X_A(2) = 1/2$ for some infinite set $A \subseteq \mathbb{N}_{>0}$.*

At each available depth M, choose the member with terminal carry 1. Its scaled terminal error is 2^{-M}, so Theorem D.2 applies along unbounded depths. The other shared-prefix conditions are not used in this compactness step. No construction at unbounded depths is established here.

The tests above leave three kinds of arithmetic information to be supplied: returns of the actual greedy sequence, finite approximants at unbounded depths, or an exclusion of a possible final skip. The following questions specify those missing inputs.

Problem D.4 (membership of $1/21$ in the Mersenne achievement set)**.** For the greedy remainder of $1/21$, prove that arbitrarily large N satisfy

$$2^{N+1} r_N(1/21) < \frac{2^{N+1}}{2^{N+1} - 1}, \quad \text{equivalently} \quad r_N(1/21) < \frac{1}{2^{N+1} - 1}.$$

Equivalently, exclude the condition $\mathcal{F}_{21}$ defined above. One sufficient route is to rule out the eventual recurrence in Theorem D.1(3) together with $s_R > 2^R$; another is to prove $s_R \le 2^R$ at arbitrarily large ranks. Neither sufficient route is claimed to be necessary by itself.

Problem D.5 (bounded returns of the scaled remainder)**.** For each of the targets $x = 1/2$ and $x = 1/21$, does the greedy remainder $r_N(x)$ return to one bounded interval after scaling by 2^N?

$$\exists B < \infty \; \forall K \; \exists N \ge K : \qquad 2^N r_N(x) \le B.$$

Problem D.6 (arithmetic tests for the greedy sequence)**.** Can a finite-memory, 2-adic or discrepancy argument prove $s_R \le 2^R$ at arbitrarily large ranks, or rule out the eventual recurrence in Theorem D.1(3)? Such an argument must use the divisor counts of the actual greedy support. Can a bounded window of R mod 6, residues of s_R, endpoint divisor counts and the finite set of eventual skips force a decrease or a contradiction? Alternatively, can one show that these bounded-memory data cannot distinguish the actual sequence from sequences that remain above 2^R but need not come from a support?

Problem D.7 (final-skip Diophantine exclusion)**.** Let $E = \sum_{n \ge 1} (2^n - 1)^{-1}$. If $1/21 \notin \mathcal{A}$, let M be the last skipped exponent, S_M its finite skipped prefix, and

$$a_M = \frac{1}{21} + \sum_{d \in S_M} \frac{1}{2^d - 1}.$$

Write $\mathrm{gap}_M = (2^M - 1)^{-1} - R_M > 0$. Can the arithmetic restrictions on this last skipped exponent be used to prove $|E - a_M| \ge \mathrm{gap}_M$, contradicting

$$0 < a_M - E < \mathrm{gap}_M \, ?$$

E Elementary details used in the proofs

Cyclotomic prime-power fact. Let $b \geq 2$, $n \geq 2$, $\ell \mid \Phi_n(b)$ be prime and $e = v_\ell(b^n - 1)$. Then $\mathrm{ord}_{\ell^e}(b) = n$. We include the 2-adic case. For odd ℓ, put $d = \mathrm{ord}_\ell(b)$ and $s = v_\ell(b^d - 1)$. The elementary lifting identity $v_\ell(b^{dt} - 1) = s + v_\ell(t)$ follows by factoring a geometric sum when $\ell \nmid t$ and by a binomial expansion for a factor ℓ. In the factorisation $b^n - 1 = \prod_{r|n} \Phi_r(b)$, subtracting these valuations over proper divisors gives

$$v_\ell(\Phi_n(b)) = \begin{cases} s & n = d, \\ 1 & n = d\ell^j,\ j \geq 1, \\ 0 & \text{otherwise.} \end{cases}$$

If $n = d$, no smaller positive exponent gives $b^m \equiv 1 \pmod{\ell}$. If $n = d\ell^j$, the full valuation is $e = s + j$; the same lifting identity shows that the least exponent giving valuation e is $d\ell^j = n$. For $\ell = 2$, b is odd. Factoring $b^{2t} - 1$ gives

$$v_2(b^{2^j} - 1) = v_2(b - 1) + v_2(b + 1) + j - 1 \qquad (j \geq 1).$$

The only possible indices $n \geq 2$ are $n = 2^j$: $v_2(\Phi_2(b)) = v_2(b + 1)$ and $v_2(\Phi_{2^j}(b)) = 1$ for $j \geq 2$; all other indices have valuation zero. The displayed identity again makes 2^j the least exponent with the full valuation. This proves the fact. Finally $\Phi_n(b) > 1$: each primitive nth root ζ satisfies $|b - \zeta| > 1$, and their product is $\Phi_n(b)$.

Exact scalar cover cost. For $t \geq 1$, put $z = \log_2 t$. The substitution $y = 2^\alpha$ reduces the infimum defining $\Psi(t)$ to $y^z/(y - 1)$ on $1 < y \leq 2$. Its derivative has the sign of $(z - 1)y - z$. Hence

$$\Psi(t) = \begin{cases} t, & 1 \leq t \leq 4, \\ z^z/(z - 1)^{z-1}, & t > 4. \end{cases}$$

The minimum is at $\alpha = 1$ for $t \leq 4$ and in the interior for $t > 4$; the quotient diverges as $\alpha \downarrow 0$. This scalar identity does not supply a converse to the positive-cover criterion.

The cost for a finite set of divisors

For the following finite divisor sets, the lower bound is asymptotically attained. Here $K_*(F)$ is the infimum of the cost K from Section 5 over finite or countable covers of F, their majorants and exponents, and positive weights of total one. A one-set cover therefore has weight 1, without the dyadic index factor in (V). For $F(q, P) = \{qd : d \mid \prod_{p \in P} p\}$, where $q \geq 2$ and no $p \in P$ divides q, put $S = \sum_{p \in P} 1/p$. If $S \geq 1$, then

$$\frac{e(S - 1)}{q} \leq K_*(F(q, P)) \leq \frac{eS}{q}. \tag{13}$$

For the lower bound, we condition on $q \mid n$ and write $f_F(n) = 2^Z$. The Chinese remainder theorem gives $\mathbb{E}Z = S$. For $z \geq 0$ and $v = \alpha \log 2 > 0$,

$$\frac{2^{\alpha z}}{2^\alpha - 1} \geq \frac{e^{(z-1)v}}{v} \geq e(z - 1) \quad (z > 1);$$

for $0 \le z \le 1$ the claimed lower bound is nonpositive. Thus $\Psi(2^z) \ge e(z-1)$; averaging and multiplying by the density $1/q$ gives the lower bound. For the upper bound, use the one-set cover with $z = 1/S$ and $\alpha = \log_2(1+z)$; its exact positive expansion has cost

$$\frac{1}{qz}\prod_{p\in P}(1+z/p) \le \frac{eS}{q}.$$

In particular, $1 - 1/S \le qK_*(F(q,P))/(eS) \le 1$. Thus $K_*(F(q,P)) \sim eS/q$ as $S \to \infty$, uniformly over the permitted choices of q and P.

F Sources, verification and reproducibility

The verification concordance lists the formal proofs by statement. *Lean* links to the supporting declarations; a dagger identifies a proof that assumes a named input. *Comparator* links to a recorded kernel check against a separately written statement; *pending* means that this comparison has not been recorded. The verification record gives the precise correspondence, dependencies and reproducible checks. A row with only a record link has no complete formal proof recorded.

For Erdős's reciprocal-summable extension and fractional-part argument, see pp. 222 and 226 of [7]. The periodic theorem [12] is also stated in Luca and Tachiya's account [11, Theorem A and Example 2, pp. 139–140], and Hornich's strict-tail theorem is proved in Nitecki's exposition [10, Theorem 4(1)]. These expositions are the sources used here for those two results; the original articles were not independently retrieved. The *Formal Conjectures* file [6] is statement-level prior art, not a proof dependency.

Formal bindings concern statements, not every printed argument. The mixed implication has a formal proof; a separate formal construction of a separating support V uses finite sets of squarefree divisors. The $A_\star$ calculation, printed fresh-prime construction, Corollary 5.2, signed finite-denominator extension and Theorem D.1 have ordinary arguments here, without independent human review. The strictness corollary has no formal binding in the record. The companion paper locates the separate comparisons for its three logarithmic-sampling results. No historical priority is asserted for the support comparison.

Reproducing the weighted theorem. The two declarations for Theorem 1.2 are in the public source at commit `91ca3405`. That snapshot pins Lean 4.29.1 in `lean-toolchain` and its Mathlib revision in `lake-manifest.json`. From a complete checkout, with `elan` installed, build the two modules with:

```
lake exe cache get
python3 scripts/lean_fast_build.py --jobs 2 \
  ErdosProblems.Erdos257.PaperCompleteR8.WeightedReturn \
  ErdosProblems.Erdos257.PaperCompleteR8.WeightedHereditaryClaim
```

The cache download is optional. The modules contain `divisibilityWeightedClaim` and `finitePrimeWeighted_fixedBase_hereditary`, respectively; their precise statements and associated checks are linked above.

The supplementary declarations are collected in the companion paper's final source section, grouped by support criteria, finite denominators, integer recurrences and achievement sets.

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Verification concordance

Statement	Formal proof	Statement	Formal proof
Theorem 1.2	Lean · Comparator	Theorem 3.1	Lean · Comparator
Theorem 4.1	Lean · Comparator	Proposition 5.1	Lean · Comparator pending
Corollary 5.2	Verification record	Theorem A.1	Lean · Comparator
Theorem B.1	Lean · Comparator	Theorem C.1	Lean · Comparator
Theorem D.1	Lean · Comparator	Theorem D.2	Lean · Comparator pending
Theorem D.3	Lean · Comparator		